

5-1-1: Examining Network Commands in Windows

At the end of this episode, I will be able to:

1. Identify and explain common network commands in Windows systems, given a scenario.

Learner Objective: *Identify and explain common network commands in Windows systems, given a scenario*

Description: In this episode, the learner will examine common network troubleshooting commands. We will explore command such as ping, ipconfig, nslookup, tracert, netstat and more.

- What commands can be used to troubleshoot the network?
 - o *ipconfig* - various utilities to display and configure TCP/IP settings in Windows (ipconfig) and Linux (ip/ifconfig/)

```
ipconfig (displays IP configuration information)
ipconfig /all (displays detailed IP configuration information)
ipconfig /release (discontinues use of currently DHCP-assigned IP address and op
ipconfig /renew (initiates the DHCP lease process D.O.R.A)
```

- o *ping* - tests reachability of and endpoint or host

```
ping 192.168.0.10 - sends ICMP echo requests to 192.168.0.10 to check connectivi
ping -n 2 www.google.com - "-n" specifies number of ICMP echo requests to send
```

- o Process

- ♠ Start with 127.0.0.1 (Loopback, verifies TCP/IP is functional)
- ♠ Next, ping the local host's IP address (verifies local IP address)
- ♠ Next, ping another host on the local network (verifies communication on the local network)
- ♠ Next, ping the default gateway (verifies communication with the router, connected to the local network)
- ♠ Finally, ping a remote host (verifies connectivity beyond the local network)

- What if we wanted to determine where these packets are going, what can we use?
 - o *tracert* - network path tracing

```
tracert www.google.com (traces the route/path taken to destination in Windows-ba
```

- With name resolution being so important to networking, how about a command that can help with DNS?
 - o *nslookup* - tests DNS name resolution

```
nslookup <domain>
nslookup acilearning.com (perform name lookup)
nslookup acilearning.com 8.8.8.8 (perform name lookup with DNS server 8.8.8.8)
nslookup <press enter> (Enters interactive mode)
  set type=AAAA (sets lookup type to IPv6)
  google.com (performs name lookup for AAAA records)
  set type=SOA (sets lookup type to Start of Authority records)
  google.com (performs lookup for SOA records)
  exit
```

- o *ipconfig*

```
ipconfig /displaydns (displays local resolver cache)
ipconfig /flushdns (clears local resolver cache)
```

- If we need to troubleshoot connections and gather protocol information, does Windows have something?

- o *netstat* - displays various network statistics, including TCP connections/ports

```
netstat -a (displays the current state of all connections)
netstat -P TCP (displays network statistics for the TCP protocol)
netstat -s -e (view the statistic of Ethernet connections)
```

- How about gathering information such as routing and ARP tables?

- o *arp* - displays and modifies the local Address Resolution Protocol (ARP) table

```
arp -a (displays ARP table entries)
arp -s xxx.xxx.xxx aa-bb-cc-11-22-33 (adds a ARP entries to the table)
```

- o *route* - displays and manipulates the routing table `` route print - display full routing table route print -4/-6 - displays IPv4 or IPv6 routing table

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- Additional Resources:

- o If applicable